

Detecting Periodically Collapsing Bubbles in the S&P 500

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Highlights:

- Cointegration tests controlling for skewness and excess kurtosis on price and dividend data from the S&P 500 show evidence of periodically collapsing bubbles.
- Application of a rolling windows test (GSADF) for explosive behavior also shows evidence of bubbles.
- The GSADF test also dates bubbles. Episodes around the Spanish Flu in 1917 and the housing crisis in 2008 were detected for every relevant choice of sample and series.

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Abstract: Two robust tests on S&P 500 data show evidence of bubbles. A test of stationarity controlling for skewness and excess kurtosis is designed to detect periodically collapsing rational bubbles. Test for cointegration and direct tests on the price-dividend and price-earnings ratios cannot reject non-stationarity for all the samples considered. A test for explosive behavior on rolling window of the data also provide evidence of multiple bubbles and allow for dating of bubbles using the same set of variables. The two bubble episodes detected in all relevant cases are associated with the Spanish Flu in 1917 and the housing crisis in 2008.

Keywords: SADF, GSADF, cointegration, stock market bubble, periodically collapsing bubbles

JEL codes: C2, G1

I. Introduction

Identifying bubbles, when stock prices deviate from fundamentals, is important for both investors and policymakers but is econometrically difficult. Proper determination of the null and the alternative hypothesis requires an assumption about the type of bubble being detected and a choice of econometric features to detect. The present work focuses on the Periodically Collapsing Rational Bubble (PCRB) model of Evans (1991) for data from the S&P 500 stock market.

This study revolves around two of the most reliable tests for bubbles, the Taylor and Peel (1998) residuals augmented Dickey-Fuller (RADF) test to determine stationarity and the Phillips, Shi and Yu (2015) generalized supremum augmented Dickey-Fuller (GSADF) test for explosive behavior. The latter test is capable of dating bubble episodes as well. The tests are conducted on four primary series, the price-dividend ratio, the price-earnings ratio, and the residuals from cointegration regressions of dividends and earnings on the price.

The evidence for the presence of bubbles is overwhelming, though the dating of bubble episodes depends on the choice of test variable and sample. The periods that show bubble behavior for all the tests are associated with the Spanish Flu and Post World War I in 1917 and the financial crisis in 2008 that started in the housing market.

Taylor and Peel (1998) show that testing for cointegration with standard unit root tests is biased in the presence of PCRBs, and they develop a test that controls for the potential skewness and excess kurtosis. The Phillips, Shi and Yu (2015) test is based on the work of Phillips, Wu and Yu (2011) who develop a test for explosive behavior based on ADF tests for a rolling window of samples. Phillips, Shi and Yu (2015) show that this sup ADF (SADF) technique only

works well when there is single bubble in the series. The generalized sup ADF (GSADF) test is an extension of SADF test which is more effective for multiple periodically collapsing bubbles. Instead of fixing starting point at the first observation in the full data, GSADF employs flexible starting and ending points. Homm and Breitung (2012) discuss the SADF test in the context of other related econometric approaches.

Our goal is to provide a complete set of results using reliable techniques on updated S&P 500 data with a number of plausible variables. The techniques are the RADF test of Taylor and Peel (1998) and the GSADF test of Phillips et al. (2015). The variables include the price-dividend ratio, the price-earnings ratio and cointegration residuals for both pairs. This paper extends the sample, which is desirable since most studies do not include the end of the Great Recession collapse.

II. Literature Review

There are a number of earlier studies that focus on the S&P 500 data through the 1980s using cointegration and unit root tests for bubbles. The results of those papers either reject the presence of bubbles, such as Taylor and Peel (1998) or Campbell and Shiller (1987), or give mixed results. Froot and Obstfeld (1991) find evidence of a non-linear relationship between price and dividends but interpreting that as evidence of a bubble has difficulties. Diba and Grossman (1988) use a cointegration test and find mixed results. Craine (1993) reports similar results for cointegration tests that allow for a time-varying discount factor. Cunado, Alana and Gracia (2005) estimate fraction unit root processes to detect deviations from long run equilibrium in price and dividend data for NASDAQ, whereas the cointegration approaches described above restrict attention to orders of integration of one or zero.

There are alternative approaches to test for bubbles. Hall, Psaradakis and Sola (1999) use a regime shifting approach to identify changes between stationary and integrated behavior. They identify deviations from long run equilibrium for prices and dividends but state that bubbles are not the cause. A related approach in Busetti and Taylor (2004), which is discussed in Homm and Breitung (2012), uses a Chow test to detect such a switch. Their Monte Carlo simulations suggest that the modified test of Busetti and Taylor (2004), and Chow test outperform the SADF test of Phillips, Wu and Yu (2011) if there is one crash and when the crash occurs at the end of sample. However, it does not hold when there are multiple crashes.

Phillips, Shi and Yu (2015) introduce the rolling GSADF tests that allow for multiple bubbles, building on the test SADF test for single bubbles in Phillips, Yu and Wu (2011), and show evidence for a number of bubbles episodes in the sample 1871-2010. The present paper uses the same test for extended sample and selection of variables. Our results largely agree, but some bubble episodes that arise for Phillips, Shi and Yu (2015) are not apparent for all the tests in the present paper, the technology bubble around 2000 being a prominent example.

Bala (2016) also applies the GSADF technology to analyze the price-dividend ratio for a smaller sample and identifies several bubbles: 1974, Black Monday 1986-1988, the dotcom bubble 1996-2001, and the housing crisis bubble 2008-2009.

The SADF test has been applied to agricultural commodity markets by Gutierrez (2011) finding bubble episodes. Bohl, Kaufman and Stephan (2013) applies SADF test and the regime shifting test of Hall, Psaradakis and Sola (1999) on daily and weekly data of ÖkoDAX and the DAX subsector Renewable Energies. Their tests exhibit speculative bubbles in both series and identify starting and ending points. Zhenya et al. (2016) reports the SADF and GSADF tests on

monthly data of Shanghai price-dividend ratio. The SADF test identifies one bubble whereas the GSADF tests identifies three periodically collapsing bubbles between 2000-2015.

This work focuses on the PCR model of Evans (1991), which is most common for empirical work, though other bubble models exist. The original rational bubble model is that of Blanchard and Watson (1982). Parke and Waters (2014) and Waters (2018) develop a model where agents switch between forecasting strategies, including a rational bubble forecast.

Other bubble models focus on learning key parameters. Lansing (2010) describes how agents update a key parameter in the intrinsic bubble model of Froot and Obstfeld (1991) to generate observed levels of asset price volatility. Adam, Marcet and Niccolini (2016) develop a consumption-based asset pricing model where the subjective estimate of the discount factor is updated according to a learning mechanism. The model can match a number of empirical features of the S&P 500 data.

III. Theoretical Framework

A standard asset pricing model demonstrates the possibility of rational bubbles and the relationship with cointegration between the price and dividends. The model of periodically collapsing bubbles is the primary alternative hypothesis to the null of cointegration for those tests and shows the potential complications. It also gives an example of the explosive behavior that the Phillips, Shi and Yu (2015) try to detect.

Let p_t be the log price of asset measured at period t , d_{t+1} is the log dividend entitled for holding stock from period t to $t+1$, the basic pricing equation is:

$$p_t = (1 + r)^{-1} E_t(p_{t+1} + d_{t+1}) \quad (3.1)$$

where $(1 + r)^{-1}$ is the discounted factor and $0 < (1 + r)^{-1} < 1$. The variable d_{t+1} represents a fundamental value such as dividends or earnings. The general solution for (3.1) is

$$p_t = \sum_{j=1}^{\infty} (1 + r)^{-j} E_t d_{t+j} + b_t \quad (3.2)$$

where $p_t^f = \sum_{j=1}^{\infty} (1 + r)^{-j} E_t d_{t+j}$ is the fundamental stock price associated with the strong efficient markets hypothesis, so equation (3.2) can be written

$$p_t = p_t^f + b_t.$$

The term bubbles b_t must satisfy the sub-martingale property

$$E_t b_{t+1} = (1 + r) b_t \quad (3.3).$$

Blanchard and Watson (1982) were the first to specify a process satisfying equation (3.3) that represents a rational bubble. One can argue that the bubble term must be zero due to a transversality condition, since asset prices cannot diverge indefinitely, but the models of Evans (1991) and Waters (2018) give examples of bubbles that rise only temporarily, so they cannot be ruled out. Equation (3.2) can be written

$$p_t - \frac{1}{r} d_t = \left(\frac{1 + r}{r} \right) \sum_{j=1}^{\infty} \left(\frac{1}{1 + r} \right)^j E_t \Delta d_{t+j} + b_t$$

For standard specifications of the dividend process, differenced dividends Δd_t is stationary, so if there is no bubble $b_t = 0$, then the price and dividend must be cointegrated.

Rejection of cointegration between p_t and d_t implies the lack of a common trend and constitutes evidence of a bubble. Diba and Grossman (1988) argue that the one-sided test of Bhargava is

superior to the standard Dickey-Fuller tests for asset markets and develop test statistics for such tests. However, Evans (1991) demonstrated such tests are insufficient to detect rational bubbles that collapse to a nonzero value and then increase again. The model of periodically collapsing bubbles is given by the following:

$$b_{t+1} = (1 + r)b_t \varepsilon_{B,t+1} \quad \text{if } b_t \leq \bar{b} \quad (3.5a)$$

$$b_{t+1} = [\xi + \pi^{-1}(1 + r)\theta_{t+1}(b_t - (1 + r)^{-1}\xi)]\varepsilon_{B,t+1} \quad \text{if } b_t > \bar{b} \quad (3.5b)$$

Where $1 + r > 1$ and $\varepsilon_{B,t} = \exp\left(y_t - \frac{\tau^2}{2}\right)$ with $y_t \sim N(0, \tau^2)$, θ_t measures the probability with Bernoulli process. θ_t takes the value of 1 with probability π and zero with probability $(1 - \pi)$, where $0 < \pi \leq 1$. When bubbles size is smaller than threshold ($b_t \leq \bar{b}$), it grows with rate $(1+r)$. After bubbles cross threshold ($b_t > \bar{b}$), it grows faster with the rate $\pi^{-1}(1 + r)$. The bubbles can also collapse with probability $(1 - \pi)$ to nonzero value ξ and the process then resumes.

The PCRB model satisfies rational expectations and does not violate a transversality condition. The bubble is rational in the sense that there are no excess returns $E_t b_{t+1} = (1 + r)b_t$. Since bubbles collapse, the price does not diverge in the long run. The model also gives an intuitive description of speculative behavior, since agents try to take advantage of the explosive behavior even with the knowledge of an eventual collapse, see Waters (2019).

Taylor and Peel (1998) test for cointegration based on equation (3.3), but studies such as Phillips, Wu and Yu (2011) and Phillips, Shi and Yu (2015) take the extra step of log linearizing the model (3.2), following the work of Campbell and Shiller (1988), where the price dividend ratio is expressed as follows for a constant c and a rate r' determined by the linearization.

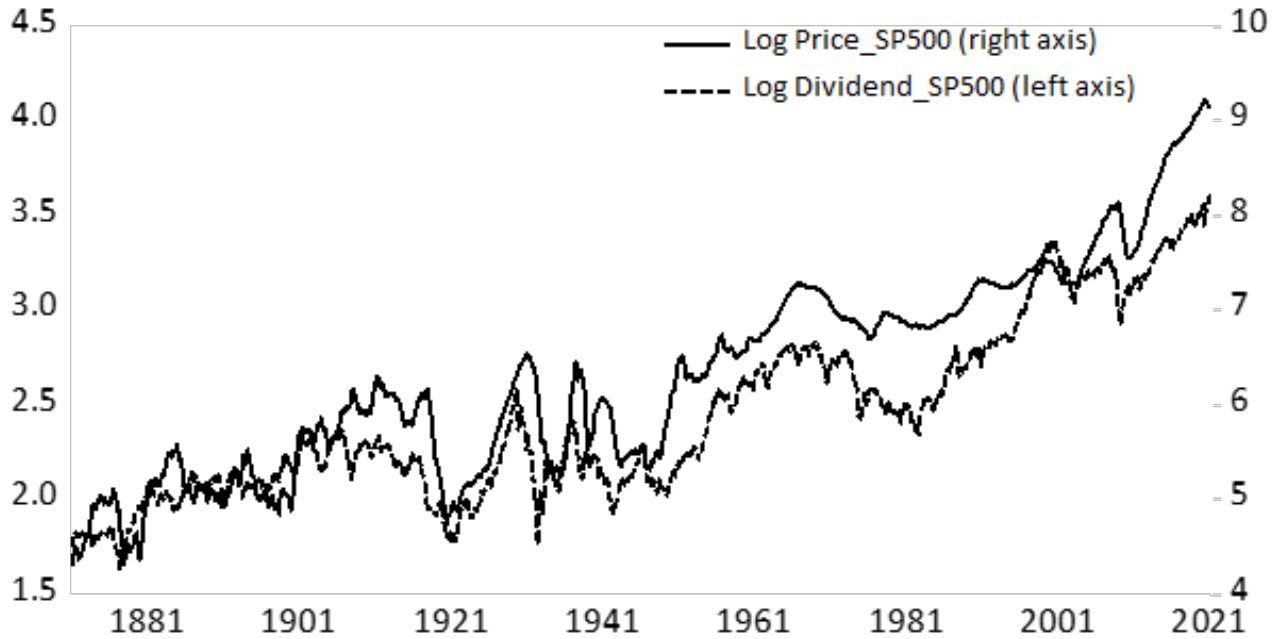
$$p_t - d_t = c + \sum_{j=1}^{\infty} \left(\frac{1}{1+r'} \right)^j E_t \Delta d_{t+j} + b_t \quad (3.4)$$

They test the price-dividend ratio for explosive behavior shown by the bubble term specification (3.3). The present work also tests for stationarity of the price-dividend and price-earnings ratio, where the null hypothesis is any type of non-stationarity.

IV. Data and Preliminary Estimations

The S&P500 real price and real dividend data are obtained from Robert Shiller's website for the period 1871-2020 and includes 1800 monthly observations. Figure 4.1 is the plot of S&P 500 log real price and log real dividend.

Figure 1: S&P 500 Log Price and Log Dividends: Monthly Data January 1870 - December 2020



Standard cointegration tests reject the presence of bubbles for any of the subsamples considered. These tests involve checking to see if the price and dividend are integrated of the same order, then testing for a unit root in the residuals of a least square regression on those two variables. Besides the *full sample* 1871m01-2020m12, the *post-war sample* 1947m01-2020m12 (Post WWII) is considered, following a number of empirical finance studies. We also consider the *hybrid sample* 1871m01-1987m12 to allow for direct comparison with the results of Taylor and Peel (1998).

To see if the price and dividend are integrated of the same order, we use the standard Augmented Dickey-Fuller Test (ADF) and Phillips-Perron (PP) test. Tables 2a-2c shows the results for unit root test in level and the first difference of the S&P 500 log real price, log real dividend and log real earning for the different samples².

Table 2a: Full Sample Jan 1871- Dec 2020

	p_t	Δp_t	d_t	Δd_t	e_t	Δe_t
ADF	-0.24 (2)	-32.27*** (0)	-0.67 (4)	-13.00*** (3)	-2.44(13)	-10.76***(15)
PP	-0.37	-32.54***	-0.61	-27.43***	-2.03	-16.06***

Table 2b: Truncated sample Jan 1871- Dec 1987

	p_t	Δp_t	d_t	Δd_t	e_t	Δe_t
ADF	-2.04 (1)	-28.06*** (0)	-2.01 (2)	-15.42***(1)	-2.39(2)	-12.03***(1)
PP	-2.00	-28.21***	-1.97	-24.46***	-2.02	-16.02***

Table 2c: Post WWII sample Jan 1947 - Dec 2020

	p_t	Δp_t	d_t	Δd_t	e_t	Δe_t
ADF	-0.51 (1)	-23.49*** (0)	-0.76 (1)	-11.00***(0)	-	-8.30***(15)
PP	-0.54	-23.63***	-0.75	-10.74***	2.71*(16)	-10.62***

The values are stationary tests *t*-statistics. The number inside the parenthesis is the lag length for ADF tests.

² Throughout the paper, significance at 1% level is denoted by (***), 5% by (**) and 10% (*).

The results indicated in Table 2a-c show that S&P 500 log real price and log real dividend are integrated of order one $I(1)$ for all the samples. The results for earnings are similar but somewhat less conclusive. Differenced earnings are clearly stationary, but the results for the unit root in earnings have p -values below 0.2, and less than 0.1 for the post-war sample for both tests. There is a large difference between the level and the difference, so we still conclude that earnings are $I(1)$.

The next step is to conduct a least squares regression

$$p_t = \alpha + \beta d_t + \varepsilon_t^d \quad (2a)$$

of the dividend d_t and a constant α on the price p_t , then check whether the residuals $\hat{\varepsilon}_t^d$ are stationary. For earning as a fundamental, the regression equation is analogous.

$$p_t = \alpha + \beta e_t + \varepsilon_t^e \quad (2b)$$

The estimated coefficient $\hat{\beta}$ on dividends for the full, hybrid and post war samples is 1.59, 1.29 and 1.84, respectively, so the choice of sample does make a difference in the cointegrating relationship. The results on the residuals are shown in Tables 3a-c.

Table 3a: Full sample Jan1871- Dec 2020.

	$\hat{\varepsilon}_t^d$	$\hat{\varepsilon}_t^e$	$p_t - d_t$	$p_t - e_t$
ADF	-4.60*** (1)	-5.00***(1)	-2.22(2)	-4.33***(1)
PP	-4.59***	-4.57***	-2.37	-3.96***

Table 3b: Truncated sample Jan 1871- Dec 1987

	$\hat{\varepsilon}_t^d$	$\hat{\varepsilon}_t^e$	$p_t - d_t$	$p_t - e_t$
ADF	-4.93*** (1)	-4.28***(1)	-4.22***(1)	-4.36***(1)
PP	-4.86***	-4.06***	-4.01***	-4.12***

Table 3c Post WWII sample Jan 1947 – Dec. 2020

	$\hat{\varepsilon}_t^d$	$\hat{\varepsilon}_t^e$	$p_t - d_t$	$p_t - e_t$
ADF	-2.63*** (1)	-4.71(3)***	-1.38(1)	-3.44***(3)
PP	-2.63***	-3.44***	-1.37	-2.60*

The values are stationary tests t-statistics. The number inside the parenthesis is the lag length for ADF tests.

The results presented in Tables 3a-3c show that the error term series is stationary at 1% significant value for all three samples and both tests on both dividends and earnings, which is evidence against explosive behavior.

Interpreting the asset pricing model as a linearization (3.4), testing for stationarity of the price-dividend and price earnings ratio is another method of detecting bubbles. The results for the former do not reject non-stationarity, in contrast to the cointegration tests. In contrast, the price earnings ratio does appear to be stationary, which is a warning the dividend and earnings series are not equivalent.

Except for the price-dividend ratio, there is no evidence for bubbles from the standard unit root tests. However, as Evans (1991) noted, these tests can have trouble identifying periodically collapsing bubbles, discussed in section 2.

V. Methodology and Results

1. Residuals Augmented Dickey-Fuller (RADF) Test

In response to Evans' (1991) concerns about standard unit root tests, Taylor and Peel (1998) suggest a residuals augmented Dickey-Fuller (RADF) test for stationarity, which controls for the skewness and excess kurtosis that might be present under periodically collapsing bubbles. The RADF estimation equation for the stationarity of the residuals ε_t from equations (2a) or (2b) is

$$\Delta \varepsilon_t = \theta \varepsilon_{t-1} + \gamma_1 \widehat{w}_{1,t} + \gamma_2 \widehat{w}_{2,t} + \mathfrak{I}_t \quad (4),$$

where the vector $\widehat{w}_t = [(\widehat{\varepsilon}_t^3 - \widehat{\sigma}^2 \widehat{\varepsilon}_t), (\widehat{\varepsilon}_t^2 - \widehat{\sigma}^2)]$ is constructed using the estimated mean $\widehat{\varepsilon}_t$ and variance $\widehat{\sigma}^2$ of the residuals ε_t . Without the skewness and excess kurtosis terms $\widehat{w}_{1,t}$ and $\widehat{w}_{2,t}$, respectively, the test reduces to a Dickey-Fuller (DF) test, the ADF test with zero lags. For all of these tests, the null is $\theta > 0$, test statistics takes the form $\frac{\widehat{\theta}}{\sqrt{\text{var}(\widehat{\theta})}}$, and the distribution is

non-standard, so Monte Carlo experiments are necessary.

Table 4 shows the results for the RADF tests on the residuals of (2a) and (2b) along with the price-dividend and price-earnings ratios.

Table 4: RADF test

Sample	$\widehat{\varepsilon}_t^d$	$\widehat{\varepsilon}_t^e$	$p_t - d_t$	$p_t - e_t$
Full (Jan 1871 – Dec 2020)	-3.0578 (0.1167)	-1.934 (0.528)	-1.294 (0.626)	-1.338 (0.612)
Truncated (Jan 1871 – Dec 1987)	-3.0227 (0.12)	-2.274 (0.409)	-2.509 (0.114)	-2.343 (0.607)
Post WWII (Jan 1941 – Dec 2020)	-2.4779 (0.34)	-0.747 (0.876)	-1.082 (0.729)	-0.154 (0.943)

The values in this table are t-statistics with p-values inside the parenthesis. The null hypothesis is non-stationarity.

The p -values are generated via Monte Carlo experiments on simulated data, again following Taylor and Peel (1998). For example, for the first case in Table 4, price data is simulated using a unit root with drift process

$$p_t = k + p_{t-1} + \eta_t \quad (6)$$

where the constant k and standard deviation of the shocks η_t are given by the mean and standard deviation of the difference of the log real price of the S&P 500 for the full sample. Series of length 1800, with initial value p_0 , matching the data, are simulated 10,000 times. For the

cointegration tests, an analogous simulation is done for log real dividends as well. Then the simulated dividends and a constant are regressed on the simulated prices, residuals ε_t analogous to those in equation (2) for each simulation. Then the RADF tests statistics are computed, generating a distribution for each, determining the p -values.

The results in Table 4 are quite different than the ADF and PP tests in Table 3. For the more reliable RADF statistic, the least decisive result is the cointegration tests on $\hat{\varepsilon}_t^d$ for the hybrid and full sample, though neither is able to reject non-stationarity at the 10% significance level. This result is not exactly the same those in Taylor and Peel (1998), which uses annual data for the hybrid sample. They do reject non-stationarity, but only because it barely meets the 5% significance threshold for both the DF and RADF tests. For the other samples and variables in Table 4, bubbles cannot be rejected at all.

The results in Table 4b for earnings are more decisive and bubbles cannot be ruled out. For the RADF test, all the p -values are greater than 0.4 so non-stationarity cannot be rejected at any reasonable level of significance.

2. Sup augmented Dickey–Fuller (SADF) tests

Another approach is to test for the explosive behavior produced by a rational bubble. Phillips, Wu and Yu (2011) suggests the Sup augmented Dickey-Fuller (SADF) test that uses recursive regression techniques to test the unit root against the alternative right-tailed explosive hypothesis. By simulation studies, Phillips, Wu and Yu (2011) demonstrate the ability of the SADF test in detecting periodically collapsing bubbles. The GSADF test introduced by Phillips, Shi, and Yu (2015) is based on the SADF test but allows for multiple bubble episodes. Another advantage of these tests is that they allow for dating of the bubble episodes.

Both tests are different than the cointegration tests in the previous section in that they test against an alternative hypothesis of explosive behavior, which is motivated by the bubble term (3.3), where the null is a unit root, rather than testing for any variety of non-stationarity. The SADF are based on the following reduced form equation:

$$y_t = \mu + \delta y_{t-1} + \sum_{i=1}^k \varphi_i \Delta y_{t-1} + \varepsilon_t, \quad \varepsilon_t \sim N(0, \sigma^2) \quad (7)$$

estimated on a set of subsamples that adds a succeeding data point for each. The null hypothesis is $\delta = 1$, the alternative is $\delta > 1$ and the lag length J is given by the method of Campbell and Perron (1997). For a sample of size T , let the fraction $r_1, r_2 \in [0, 1]$ determine the starting and ending points of the sample, $[r_1 T]$ and $[r_2 T]$, where $r_2 > r_1$, so the size of the subsample is $n = [(r_2 - r_1)T]$.

Given a fixed r_1 , the ADF_r statistic is computed for each subsample allowing $r = r_2 \in [r_1, 1]$.

PYW (2011) show that ADF_r is distributed such that

$$ADF_r \Rightarrow \frac{\int_0^r \tilde{W} dW}{\int_0^r \tilde{W}^2}$$

$$\sup_{r \in [r_1, 1]} ADF_r \Rightarrow \sup_{r \in [r_1, 1]} \frac{\int_0^r \tilde{W} dW}{\int_0^r \tilde{W}^2} \quad r \in [r_1, 1]$$

Where the term $\tilde{W}$ is demeaned standard Brownian motion, see PYW (2011) for details and a proof.

The SADF test for a unit root compares the estimated value of $\sup_{r \in [r_1, 1]} ADF_r$ and the critical values given by $\sup_{r \in [r_1, 1]} \frac{\int_0^r \tilde{W} dW}{\int_0^r \tilde{W}^2}$ for a chosen level of significance.

To detect bubbles, we apply the SADF test on the full sample, truncated, and post war samples of S&P 500 log real price-dividend ratio, the log real price-earnings ratio, and the errors

from the cointegration regressions ε_{dt} and ε_{et} from equations (2a) and (2b). Table 5 shows the t -statistics for the test of the null of a unit root $\delta=1$ versus the alternative of $\delta>1$ in equation (7).

Table 5: SADF test for explosive behavior in S&P 500

	ε_{dt}	ε_{et}	$p_t - d_t$	$p_t - e_t$
Full Sample	0.50	0.73	0.69	0.58
Truncated	0.60	0.24	0.69	0.51
Post WWII	-0.16	3.34***	0.87	2.57***

The null hypothesis is a unit root $\delta=1$ and alternative is explosiveness $\delta>1$ in equation (7). Critical values are shown in Table 5a in the Appendix.

There is evidence for bubbles only in the earnings regression for the post-WWII sample. The results using dividends contrast with those of Phillips et.al (2015), Bala and William (2016), and Itamar (2017) who use the same test on different samples. However, when earnings is used as the fundamental, there is evidence of bubble on the post-WWII sample.

3. Generalized Sup Augmented Dickey-Fuller (GSADF) Test

As noted by Phillips, Wu and Yu (2015), SADF test may show inconsistency if there are multiple bubbles, which is a concern for the samples used here. To address the weakness of the SADF test in evaluating multiple periodically collapsing bubbles, Phillips, Shi and Yu (2011) suggest a new bubble detecting and dating technique known as the generalized sup ADF test, or GSADF. They compared the SADF model and the new model GSADF in terms of size and power, the simulation results confirmed the generalized model's supremacy. The main distinction between the two techniques is that the GSADF test varies both the starting and ending points of the recursion.

Phillips, Shi and Yu (2011) employ recursive regression techniques with the same principle as the SADF test, testing the null of $\delta=1$ for equation (7), but with a moving window

(r_1, r_2) in which the fractional starting point r_1 of the sample also varies as if the ending point r_2 . In other word, r_1 can be any value from 0 to $r_2 - r_0$. Since r_1 is also changing, the number of regressed sample increases, which increases the power of the GSADF test. The GSADF statistics is obtained by taking largest ADF statistics over the flexible range r_1 and r_2 :

$$GSADF(r_0) = \sup ADF_{r_1}^{r_2} \quad r_2 \in [r_0, 1], r_1 \in [0, r_2 - r_0]$$

Phillips, Shi and Yu (2011) propose a date-stamping technique that uses backward expanding sample sequences to determine the dates of explosive behavior. Suppose r_2 is fixed while r_1 moves in the range $[0, r_2 - r_0]$ for a chosen minimum r_0 (see PSY, p. 1050), we denote $BAADF_{r_1}^{r_2}$ which is the ADF statistics for each regression of the range from r_1 to r_2 . The $\{BAADF_{r_1}^{r_2}\}_{r_1 \in [0, r_2 - r_0]}$ is the ADF statistics sequence starting from r_1 and ending at r_2 . The $\sup\{BAADF\}$ statistic is defined as a backward looking SADF statistic.

$$BSADF_{r_2}(r_0) = \sup\{ADF_{r_1}^{r_2}\}_{r_1 \in [0, r_2 - r_0]}$$

The GSADF statistics is defined as the $\sup BSADF_{r_2}(r_0)$ for moving r_2 from r_0 to 1

$$GSADF(r_0) = \sup\{BSADF_{r_2}(r_0)\}$$

The BSADF statistics is used to estimate the origination and termination date of the bubbles. The estimated origination date $T\hat{r}_e$ is the first observation in which the BSADF statistic greater than the critical value. The estimated termination date $T\hat{r}_f$ is the first observation with a BSADF statistic less than the critical value after the date $T\hat{r}_e + \delta \log(T)$, where δ is the frequency of data.

The fractional corresponding to $T\hat{r}_e$ and $T\hat{r}_f$ are $\hat{r}_e$ and $\hat{r}_f$ which are defined as:

$$\hat{r}_e = \inf \{r_2: BSADF_{r_2}(r_0) > scv_{r_2}^{Bt}\} \quad r_2 \in [r_0, 1]$$

$$\hat{r}_f = \inf\{r_2: BSADF_{r_2}(r_0) < scv_{r_2}^{B_t}\} \quad r_2 \in \left[\hat{r}_e + \frac{\delta \log(T)}{T}, 1\right]$$

$cv_{r_2}^{B_t}$ is the righted tail critical value of ADF statistics. The significance level B_t varies with t to ensure consistent estimation.

Table 6 shows the tests for bubbles using the GSADF test. The format is the same as Table 5, which contains results from the SADF test. Critical values for Table 6 are in Table 6a in the Appendix.

Table 6: GSADF test for explosive behavior in S&P 500

	ε_{dt}	ε_{et}	$p_t - d_t$	$p_t - e_t$
Full Sample	3.20***	3.88***	3.41***	3.70***
Truncated	3.96***	3.62***		
Post War	3.51***	7.36***		

*The values presented in the table are GSADF statistics. The null hypothesis is unit root and alternative is explosiveness. Significant at 1% level is denoted by (***), 5% by (**) and 10% (*).*

In contrast to the more restrictive SADF test, explosiveness cannot be rejected for any of the tests and the t-statistics are universally large.

The tests dating bubbles verify these results, though the precise bubbles episodes identified vary somewhat with the sample and series tested. The graph in Figure 2 shows the BADF statistic (blue) against the 5% critical value for rejecting explosive behavior (red) for the price-dividend ratio on the full sample. The shaded area shows the dates when the BADF statistic exceeds the critical value, giving evidence of a bubble. Those dates are listed on Table 7. Table 8 and Figure 3 show the analogous results for the price-earnings ratio.

Table 7: Origination and termination dates on $p_t - d_t$ series on full sample

Starting date	Ending date	Duration (months)	Event
10/31/1879	04/30/1880	6	End of the Long Depression
10/31/1917	2/28/1918	4	Spanish flu
6/30/1932	7/31/1932	1	U.S. exits the gold standard
7/31/1997	11/30/1997	4	Asian financial crisis
2/28/1998	8/31/1998	6	Dot com bubble
1/31/1999	2/28/1999	1	Dot com bubble
3/31/1999	8/31/1999	5	Dot com bubble
11/30/1999	2/29/2000	3	Dot com bubble
3/31/2000	5/31/2000	2	Dot com bubble
7/31/2000	9/30/2000	2	Dot com bubble
11/30/2008	1/31/2009	2	Housing bubble
2/28/2009	4/30/2009	2	Housing bubble

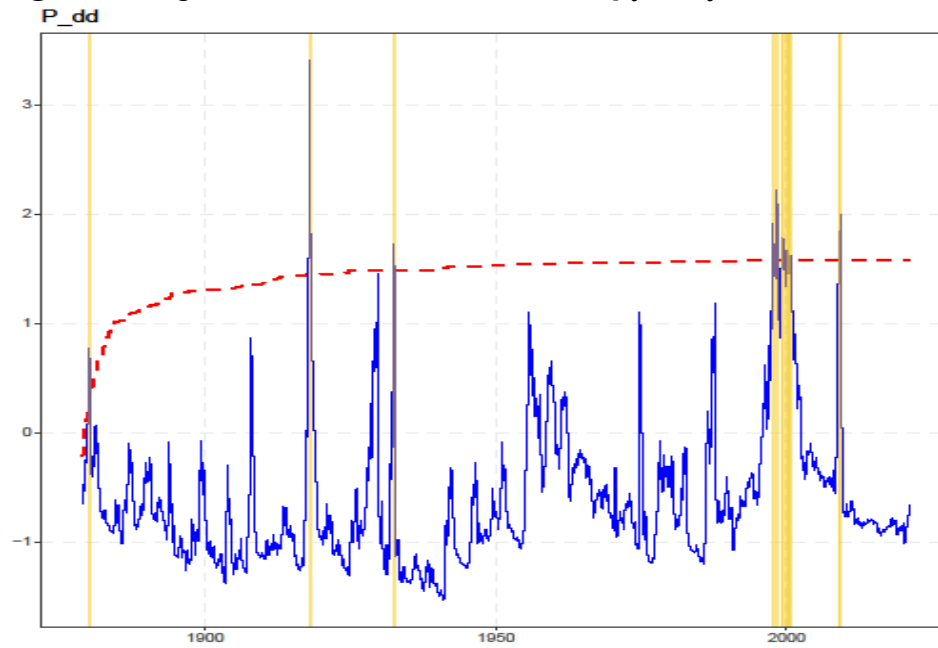
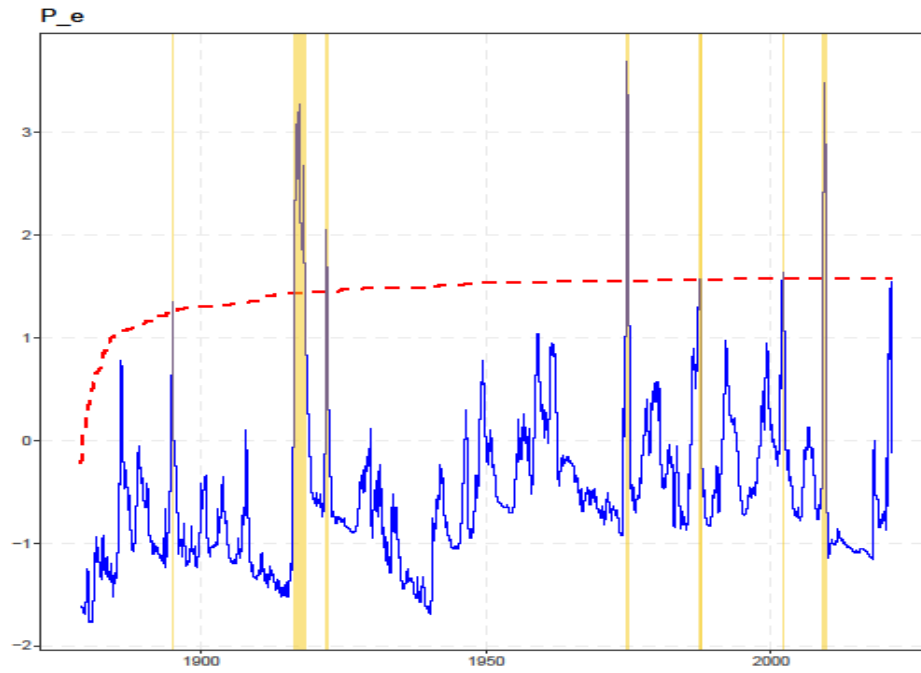
Figure 2: Origination and termination dates on $p_t - d_t$ series on full sample

Table 8: Origination and termination dates on $p_t - e_t$ series on full sample

Starting date	Ending date	Duration (months)	Event
12/31/1894	1/31/1895	1	Panic of 1893
4/30/1916	2/28/1918	22	Spanish flu
12/31/1921	1/31/1922	1	1920-1921 recession
7/31/1974	1/31/1975	6	Recession
6/30/1987	7/31/1987	1	Black Monday
8/31/1987	9/30/1987	1	Black Monday
12/31/2001	1/31/2002	1	Dotcom bubble
2/28/2009	8/31/2009	6	Housing bubble

Figure 3: Origination and termination dates on $p_t - e_t$ series on full sample



The Appendix includes the analogous Tables 9a-c of episodes and Figures 5a-c for the cointegration residuals using dividends ε_{dt} and the Tables 10a-c and Figures 6a-c for the cointegration residuals using earning ε_{et} .

The GSADF test finds presence of bubble in all series for all subsamples, though the dating of the bubbles depends to some degree on the variable estimated and the sample. Table 11 summarizes the results focusing on some of the more notable episodes detected that have a natural narrative interpretation. The historical events (loosely) associated with the dates in the table are as follows:

- 1879 – end of the Long Depression
- 1917 – Spanish Flu
- 1929 – Black Tuesday stock market crash
- 1933 – the United States abandons the gold standard
- 1974 – recession recovery
- 1987 – Black Monday stock market crash
- 1999 – Dot com bubble and crash
- 2008 – Housing bubble and mortgage meltdown
- 2020 – COVID recession

Table 11: Detection of historical episodes

Series	Sample	1879	1893	1917	1929	1933	1974	1987	1999	2008	2020
$P_t d_t$	Full	6	x	4	x	1	x	x	15	4	x
$p_t e_t$	Full	x	1	22	x	X	6	2	x	6	x
ε_{dt}	Full	4	x	4	x	3	x	x	9	6	x
ε_{dt}	Truncated	6	x	5	x	2	x	x	-	-	-
ε_{dt}	Post-WWII	-	-	-	-	-	x	x	4	6	x
ε_{et}	Full	x	2	23	x	X	6	x	x	8	1
ε_{et}	Truncated	x	x	24	1	X	6	1	-	-	-
ε_{et}	Post-WWII	-	-	-	-	-	9	3	x	10	1

The entries show the number of months with explosive behavior around historical episodes according to the GSDADF tests detailed in Tables 9-12. The 'x' symbol indicates no explosive behavior, and the '-' symbol means the episode is outside the sample.

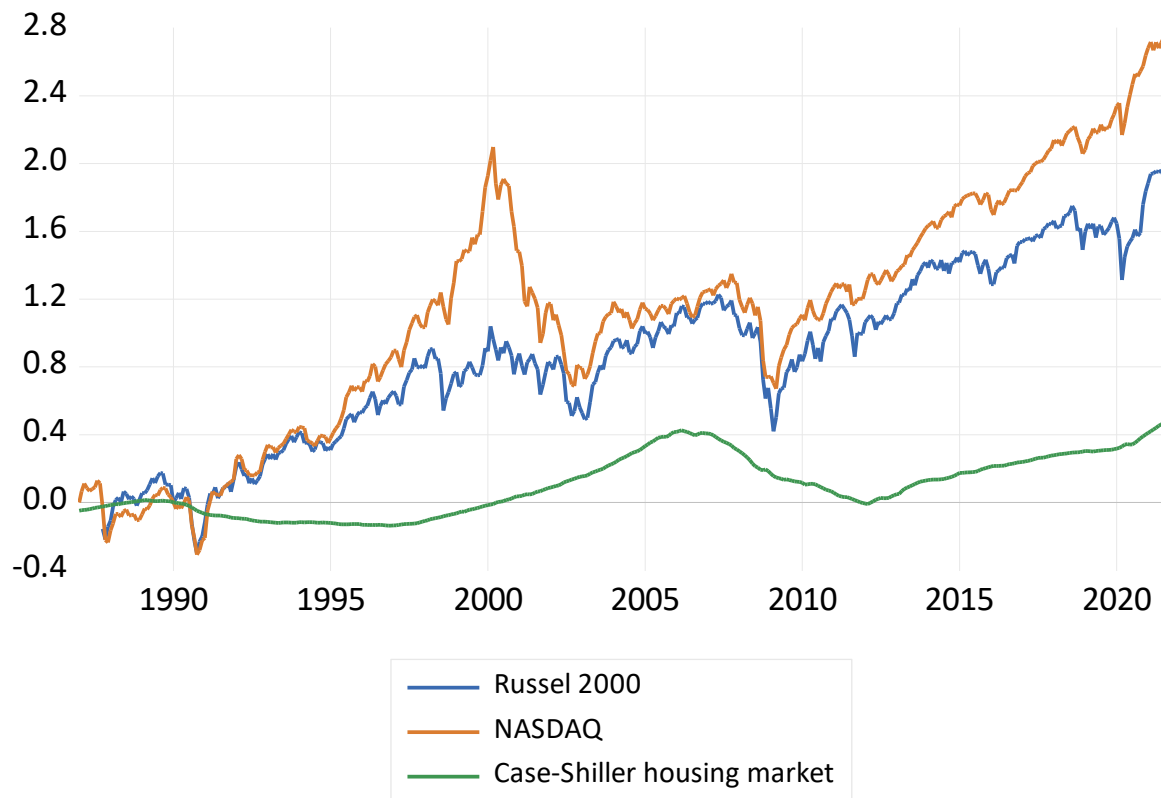
Many of the episodes of explosive behavior occur as a recovery from some type of the downtown, the partial recovery during the Great Depression in 1933 being a prime example.

Interestingly, the Black Tuesday crash leading into the Great Depression is only detected once.

The results match those of Phillips, Shi and Yu (2015), who use the sample 1871-2010, in that the tech bubble of 1999 appears prominently with the price-dividend ratio. However, the detection of the tech bubble is not confirmed by any of the tests using earnings, and the number of bubble periods detected with dividends varies substantially from 4 to 15.

To help examine the circumstances of the recent bubble episodes, Figure 4 show the log real price indices for the Russell 2000, NASDAQ and housing market given by Case-Shiller. The house price index is notably smoother than the stock indices, though the decline leading up to the housing crisis in 2008 is readily apparent. The crash in stock prices is easy to detect with the bottom occurring in February 2008 for the Russell 2000 and March 2008 for the NASDAQ.

Figure 4: Log real price indices normalized to zero in January, 1990.



The collapse of the tech bubble around 1999-2000 is less visible in the Russel 2000 index, which tracks the S&P 500 more closely. The housing market shows no change, and the NASDAQ price index peaks in March 2000, collapsing thereafter.

Figure 4 confirms the conclusion of the GSADF tests for bubbles in that there is much stronger evidence for collapse around the housing crisis in 2008 than there is for the commonly discussed tech bubble of 2000. One natural interpretation is that there was a bubble around 2000, but it did not spread beyond technology stocks. Hence, bubble tests on for the NASDAQ, as in Phillips, Wu and Yu (2001) do give evidence of bubbles, but the results on the S&P 500 presented here are less conclusive.

Though the data is limited at this time, the price indices show a very modest effect around COVID, indicating the market participants think the disruptions are temporary, and may even benefit the housing and technology sectors.

VI. Conclusion

The most reliable tests cannot reject periodically collapsing rational bubble in the S&P 500 stock index. Tests for non-stationarity and tests that detect explosive behavior on the price-dividend ratio, the price-earnings ratio and the associated cointegration residuals give similar qualitative results, though the dating of the bubbles can vary. Two events are notable since they are detected by all the relevant tests, the Spanish Flu of 1917 and the housing bubble and crash around 2008.

Preliminary GSADF tests on the most recent data indicate one period of explosive behavior in December 2021. However, the fact that the results changed from the sample ending in 2010 in the Phillip, Wu and Yu (2015) paper and the present work indicate that using the test for real time bubble detection should be done with trepidation.

Standard tests for stationarity are not capable of detecting periodically collapsing bubbles, which explains the results in Table 2-3, but the RADF test of Taylor and Peel (1998) that controls for the skewness and excess kurtosis that could arise does give evidence of bubbles in every series and sample tested. Similarly, the SADF test rejects explosive behavior, with the exception of the price-earnings ratio, but the GSADF test that allows for multiple episodes does give evidence of bubbles and also allows for dating of the episodes.

These results should give pause to investors and policymakers alike. The price data does not always reflect fundamentals and should be interpreted with great caution. The results here identify periods when markets are efficient and when they are not, which should help our

understanding of the connection between prices and fundamentals so decisions based on future market behavior can be made with more confidence.

References

- Adam, K., A. Marcet and J.P. Niccolini 2016. Stock Market Volatility and Learning. *Journal of Finance* 71(1): 33-82.
- Bala, A. and William, B.N. 2016. Testing for Stock Price Bubbles: A Review of Econometric Tools (2016). *The International Journal of Business and Finance Research* 10(4): 29-42.
- Blanchard, O. and Watson, M. 1982. Bubbles, Rational Expectations and Financial Markets. *Cambridge, Mass: National Bureau of Economic Research*.
- Bohl, M.T., Kaufmann, P. and Stephan, P.M. 2013. From Hero to Zero: Evidence of Performance Reversal and Speculative Bubbles in German Renewable Energy Stocks. *Energy Economics* 37: 40-51
- Branch, W. and G. Evans (2011). Learning about Risk and Return: A Simple Model of Bubbles and Crashes. *American Economic Review: Macroeconomics* 3: 159-191.
- Busetti, F. and Taylor, A. M. R. 2004. Tests of Stationarity Against a Change in Persistence. *Journal of Econometrics* 123: 33–66
- Campbell, J.Y. and Perron, P. Pitfalls and Opportunities: What Macroeconomists Should Know about Unit Roots. *National Bureau of Economic Research Macroeconomics Annual*: 141-201.
- Campbell, J. and Shiller, R. 1988. The Dividend-Price Ratio and Expectations of Future Dividends and Discount Factors. *Review of Financial Studies*, 1(3): 195-228.
- Caspi, I. 2017. Rtdf: Testing for Bubbles with EViews. *Journal of Statistical Software*. Code Snippet: 1 – 16.
- Craine, R. 1993. Rational Bubbles: A Test. *Journal of Economic Dynamics and Control* 17(5-6): 829-846.
- Cunado, J., Gil-Alana, L.A. and Gracia, F.P. 2004. A test for rational bubbles in the NASDAQ stock index: A fractionally integrated approach. *Journal of Banking & Finance* 29 (2005) 2633-2654.
- Diba, B. and Grossman, H. 1988. Explosive Rational Bubbles in Stock Prices? *American Economic Review* 78: 520-30.
- Evans, G. 1991. Pitfalls in Testing for Explosive Bubbles in Asset Prices. *American Economic Review* 31: 922–30.

- Froot, K. and Obstfeld, M. 1991. Intrinsic Bubbles: The Case of Stock Prices. *American Economic Review* 81: 1189-214.
- Gutierrez, L. 2011. Speculative bubbles in agricultural commodity markets. *European Review of Agricultural Economics* 40 (2): 217-238
- Hall, Detecting Periodically Collapsing Bubbles: A Markov-Switching Unit Root Test. *Journal of Applied Econometrics* 14(2): 143-154.
- Homm, U. and Breitung, J. 2011. Testing for Speculative Bubbles in Stock Markets: A Comparison of Alternative Methods. *Journal of Financial Econometrics*, forthcoming 10(1):198-231.
- Lansing, K. 2010. Rational and Near-Rational Bubbles without Drift. *Economic Journal* 120: 1149-1174.
- Parke, W. and G. Waters 2014. On the Evolutionary Stability of Rational Expectations. *Macroeconomic Dynamics* 18(7): 1581-1606.
- Phillips, P. C. B., Wu, Y. and Yu, J. 2011. Explosive Behavior in the 1990s NASDAQ: When did Exuberance Escalate Asset Values? *International Economic Review* 52: 201-26.
- Phillips, P.C.B., Shi, S.P. and Yu, J. 2015. Testing For Multiple Bubbles: Limit Theory of Real-Time Detectors. *International Economic Review* 56 (4): 1079-1134.
- Taylor, M.P. and Peel, D.A. 1998. Periodically collapsing stock price bubbles: a robust test. *Economics Letters* 61(2): 221-228.
- Waters, G. 2018. Informational Efficiency and Endogenous Rational Bubbles. *Uncertainty, Expectations and Asset Price Dynamics* edited by Jawadi, F., Springer: 149-172.
- Waters, G. 2019. Bubbles and Rationality in Bitcoin. *Economic Notes* 48(2), July.

Appendix

Table 5a: Critical Values of SADF for Explosive Alternative

	1%	5%	10%
Full Sample	2.08	1.58	1.36
Truncated	2.08	1.58	1.32
Post-WWII	2.03	1.55	1.29

The table presents the critical values of 1%, 5% and 10% significant level of SADF test based on Monte Carlo simulation. The minimum windows is 94, 81 and 63 for full sample, truncated and post war sample, respectively.

GSADF: All the tests are performed by Exuber package in R. The number of repetitions is 2000. The minimum windows are 94, 81 and 63 for full sample, truncated and post war sample, respectively which follows the formula $\{T(0.01 + 1.8/\sqrt{T})\}$ (Phillips et al., 2015) where T is the number of observations. The number of lags is fixed at zero.

Table 6a: Critical Values of GSADF for the Explosive Alternative

	1%	5%	10%
Full Sample	3.02	2.45	2.21
Truncated	3.00	2.42	2.18
Post War	2.92	2.30	2.06

The table represents the critical values of 1%, 5% and 10% significant level of GSADF; the minimum windows is 94, 81 and 63 for full sample, truncated and post war sample, respectively.

Table 9a: GSADF test origination and termination dates on ε_{dt} series on full sample

Starting date	Ending date	Duration (months)	Event
11/30/1879	3/31/1880	4	End of the Long Depression
10/31/1917	2/28/1918	4	Spanish flu
12/31/1931	1/31/1932	1	U.S. exits the gold standard
5/31/1932	7/31/1932	2	U.S. exits the gold standard
7/31/1997	8/31/1997	1	Asian financial crisis
2/28/1998	8/31/1998	6	Dot com bubble
4/30/1999	6/30/1999	2	Dot com bubble
7/31/1999	8/31/1999	1	Dot com bubble
10/31/2008	4/30/2009	6	Housing bubble

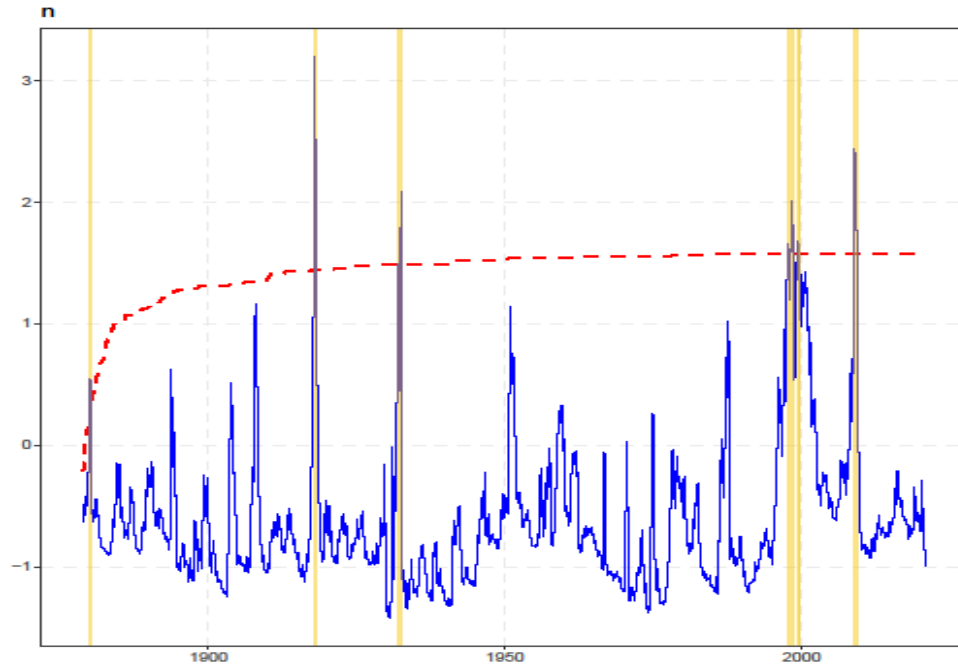
Figure 5a: GSADF test origination and termination dates on ε_{dt} series on full sample

Table 9b: GSADF test origination and termination dates on ε_{dt} series on truncated sample

Starting date	Ending date	Duration (months)	Event
10/31/1879	4/30/1880	6	End of the Long Depression
8/31/1893	9/30/1893	1	Panic of 1893
9/30/1917	2/28/1918	5	Spanish flu
5/31/1932	7/31/1932	2	U.S. exits the gold standard

Figure 5b: GSADF test origination and termination dates on ε_{dt} series on truncated sample

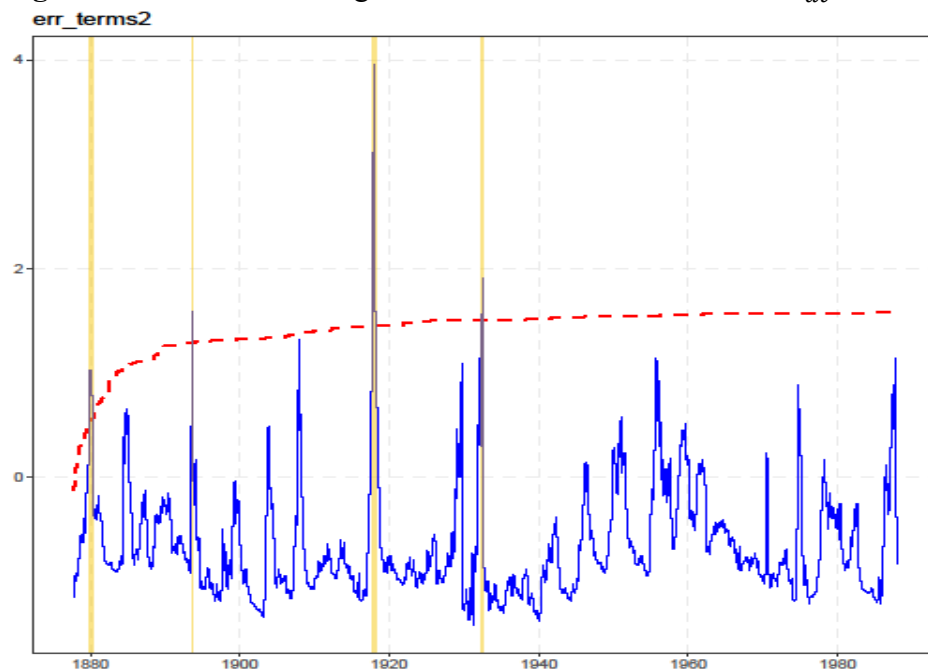


Table 9c: GSADF test origination and termination dates on ε_{dt} series for the post-WWII sample

Starting date	Ending date	Duration (months)	Event
7/31/1997	9/30/1997	2	Asian financial crisis
3/31/1998	6/30/1998	3	Dot com bubble
7/31/1998	8/31/1998	1	Dot com bubble
4/30/1999	6/30/1999	2	Dot com bubble
7/31/1999	8/31/1999	1	Dot com bubble
10/31/2008	4/30/2009	6	subprime mortgage

Figure 5c: GSADF test origination and termination dates on ε_{dt} series on post war sample

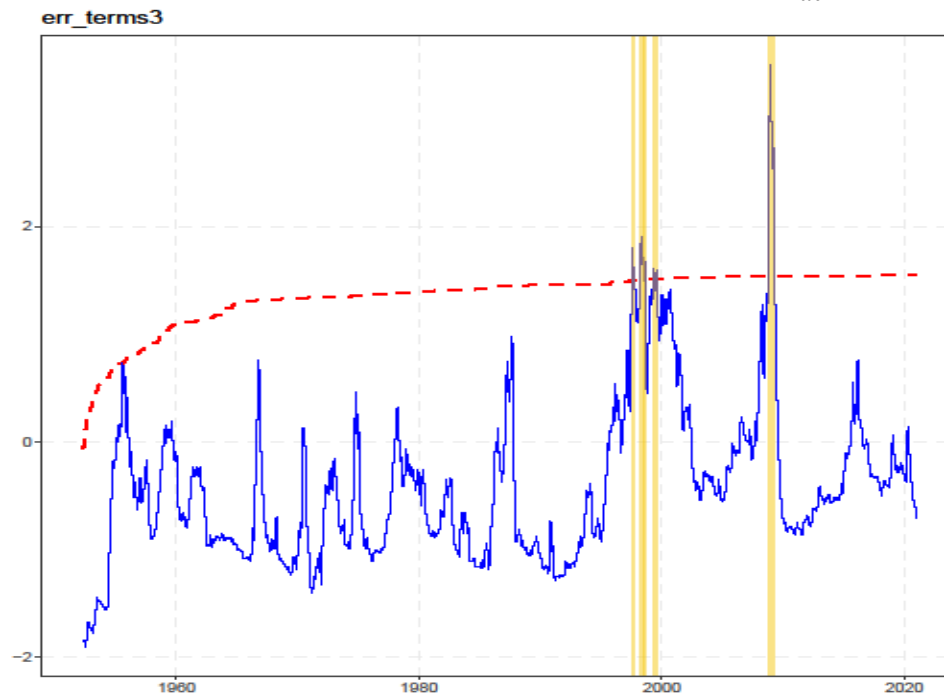


Table 10a: GSADF test origination and termination dates on ε_{et} series on full sample

Starting date	Ending date	Duration (months)	Event
11/30/1885	12/31/1885	1	Long Depression
11/30/1894	1/31/1895	2	Panic of 1893
3/31/1916	2/28/1918	23	Spanish flu
11/30/1921	2/28/1922	3	1920-1921 depression
7/31/1974	1/31/1975	6	recession
6/30/1987	7/31/1987	1	Black Monday
12/31/2001	2/28/2002	2	Dotcom bubble
3/31/2002	4/30/2002	1	Dotcom bubble
1/31/2009	9/30/2009	8	Housing bubble
12/31/2020	1/31/2021	1	Covid 19

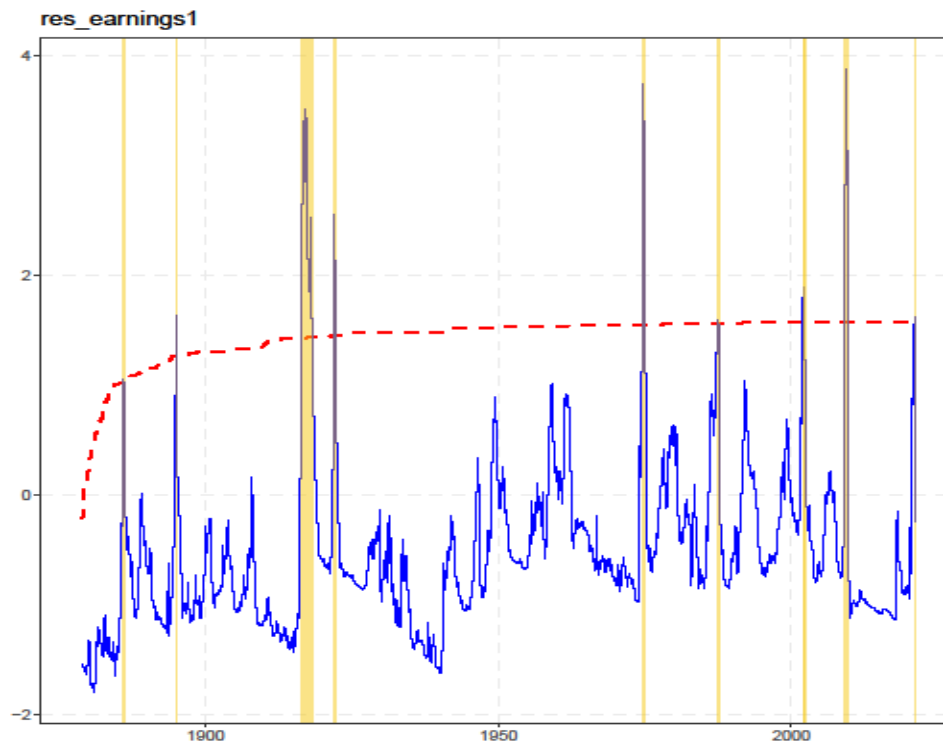
Figure 6a: GSADF test origination and termination dates on ε_{et} series on full sample

Table 10b: GSADF test origination and termination dates on ε_{et} series on truncated sample

Starting date	Ending date	Duration (months)	Event
4/30/1916	4/30/1918	24	Spanish flu
12/31/1921	1/31/1922	1	1920-1921 recession
9/30/1929	10/31/1929	1	Black Tuesday
7/31/1974	1/31/1975	6	recession
8/31/1987	9/30/1987	1	Black Monday

Figure 6b: GSADF test origination and termination dates on ε_{et} series on truncated sample

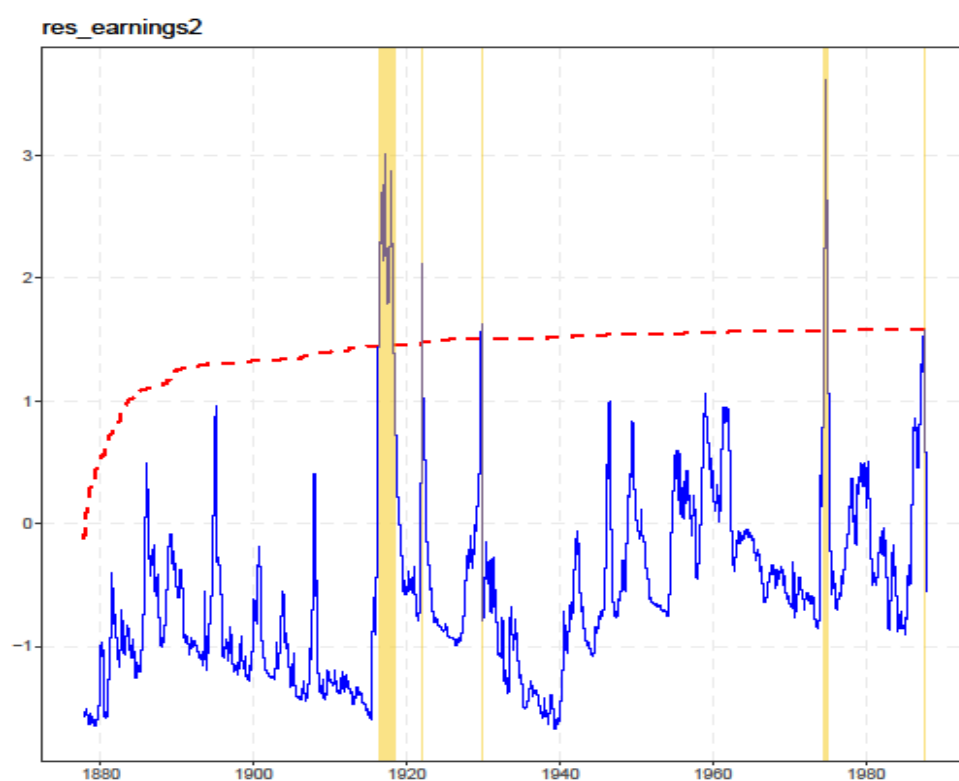


Table 10c: GSADF test origination and termination dates on ε_{et} series on post war sample

Starting date	Ending date	Duration (months)	Event
10/31/1958	2/28/1959	4	Recession
5/31/1974	2/28/1975	9	Recession
6/30/1987	9/30/1987	3	Black Monday
11/30/2001	4/30/2002	5	Dotcom bubble
11/30/2008	9/30/2009	10	Housing bubble
12/31/2020	1/31/2021	1	Covid 19

Figure 6c: GSADF test origination and termination dates on ε_{et} series on post war sample

